Braillix - Master Bill of Materials

Supersedes `docs/SHOPPING_LIST.md` (that file still lists 4mm pen springs and 2mm bearing balls - both wrong since v7.1). Buy from this file only. Scope: **1 Brain Pod + 1 Cell**, enough to make one dot go up and come back down. Budget ?15,000 - already spent ?935 (J.B. Enterprises). Fastener quantities are **derived from the** `.scad` **sources and verified**, not estimated - each row names the feature it serves. Prices are realistic Indian estimates (Amazon.in / robu.in / local market), marked **~** where not quoted from a specific listing.

Status legend

	Meaning
[OK] **HAVE IT**	Already owned - from the ?935 bill or stated in hand
[X] **NEED TO BUY**	Not owned. Where + estimated cost given
[!] **HAVE BUT WRONG**	Owned, but the wrong spec or not enough

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1. Electronics

Component	Status	Action Needed
ESP32 DevKit V1 (30-pin, USB-C)	[OK] HAVE IT	-
5V/3A power adapter	[OK] HAVE IT	-
Inline female DC pigtail jack	[OK] **HAVE FOR TESTING**	Photo-identified; no thread/nut, so it cannot mount in the pod wall. Verify centre-pin polarity by continuity before power.
Threaded 5.5x2.1mm female panel-mount jack	[X] **NEED TO BUY / SELECT**	Final enclosure part. Must have retaining nut and >=5V/3A rating; record exact drawing/link before finalizing lid CAD.
28BYJ-48 stepper motor (5V)	[OK] HAVE IT / MEASURED	Fit-critical caliper readings recorded, including M5=3.0mm. Vertical stack still needs one coordinated CAD pass.
ULN2003 driver module	[OK] HAVE IT	Fits the 16mm pocket **only** with wires soldered flat (~12mm). With vertical Dupont headers it is ~20mm and will NOT fit.
Hall sensor - MH-Sensor-Series (blue module)	[!] **HAVE BUT WRONG FORM FACTOR**	The whole blue PCB will not fit. v7.5 provides a **4.5 x 3.5 x 1.6mm underside pocket** for its bare TO-92 sensor. M11b is exactly 1.6mm, so dry-fit before gluing. Desolder the sensor and run three wires back; use analog AO behavior for homing.
Neodymium magnets 8x1mm (docking)	[OK] HAVE IT	10 purchased. 4 needed per cell (2 per +/-X face).
Homing magnet **3x2mm** (for the cam)	[X] NEED TO BUY	1 per cell. An 8x1mm will **not** fit the cam's 3.2mm pocket. Search "3x2mm neodymium magnet" - ~?120 for 10-20 pcs.
Tactile switches 6x6x5mm	[OK] HAVE IT	3 needed (pod nav buttons). Not required for a Tier-0 demo.
Female header strips 1x15	[X] NEED TO BUY	2 for the pod DevKit mount. ~?40. Not needed for a breadboard demo.
USB-C cable (data, not charge-only)	[X] NEED TO BUY / verify	Must carry **data**. Many charge-only cables look identical and silently fail to flash. ~?150.
Resistors / capacitors	[OK] **NOT NEEDED**	4.7k I2C pull-ups are only for the multi-cell I2C bus, which is not in the demo. No protection parts needed for a single cell.
Pogo pin connectors	?? **DEFERRED**	Spec: 4-pin, 2.54mm pitch, spring-loaded, ~0.5A. Not needed for a single-cell demo. Pocket dims in the CAD are placeholders pending a real part.

2. Mechanical & fasteners

Counts derived from the SCAD sources and verified against them.

Component	Qty	Status	Serves / Action
M2.5 x 25mm bolts	4	[X] NEED TO BUY	`top_plate` counterbores -> `base_plate` standoffs -> `outer_box` corner bosses. Verified: 25mm gives ~9.5mm engagement. ~?60
M4 x 10mm bolts + nuts	2	[X] NEED TO BUY	Motor mounting ears (`base_plate`, dia 4.3 holes). ~?30
M2 x 8mm self-tap	2	[X] NEED TO BUY	Pod lid -> shell bosses. [!!] **NOT M2x6** - through a 4mm lid that leaves only **2mm** of thread; 8mm gives 4mm. Verified. ~?40
M2 x 6mm self-tap	4	?? DEFERRED	Muscle-board bosses. Not used in the prototype (ULN2003 sits on the pocket floor).
2mm OD micro compression springs	6 + spares	[X] NEED TO BUY	[!!] **CRITICAL, no substitute.** 2.0mm OD, ~0.3mm stainless wire, ~4mm free length. Pen springs (4mm) **cannot** work - braille rows are 2.6mm apart. Buy an **assortment kit** (200-400pcs) so you're not betting on one guess. ~?500

3. Consumables

Component	Status	Action Needed
Solder wire	[X] NEED TO BUY	**0.8mm, 60/40 rosin-core.** Not plumbing solder. ~?150
Flux paste	[X] NEED TO BUY	Not strictly required, but makes soldering to ULN2003 pins far easier for a beginner. ~?100
Superglue (CA) **or** 5-min epoxy	[X] NEED TO BUY	Glues the resin `dot_insert` into the PETG top plate, and the magnets. **Epoxy recommended** - it gives you time to seat the tile square. ~?80
Heat-shrink tubing (1-3mm assorted)	[X] NEED TO BUY	Insulating soldered joints. ~?80
Electrical tape	[X] NEED TO BUY	~?30
Silicone grease	[X] NEED TO BUY	Cam tracks, to reduce stall risk. A tiny amount. ~?80
Dupont jumper wires	[OK] HAVE IT	2m wire + spares purchased. For counts see below.
Breadboard	[X] NEED TO BUY	**Yes, still needed** - the Tier-0 demo is breadboard-based, and it is the safest way to bring up wiring before anything is soldered. Half+ size. ~?100

Dupont counts for the demo wiring (13 connections total): **M-F x8** (ESP32 header -> breadboard/module), **M-M x5** (rail to rail, module to rail). F-F not needed. If your 2m wire is bare, you need crimps or you solder directly.

4. Tools

Tool	Status	Action Needed
Soldering iron	[OK] **HAVE IT**	You stated basic experience and an iron. 25-60W with a conical/chisel tip is right.

Tool	Status	Action Needed
Digital multimeter	[X] **NEED TO BUY - MANDATORY**	[!!] The single most important purchase. Barrel-jack polarity is unverified and getting it wrong destroys the ESP32 on first power-up. Also your only way to debug a dead rail. ~?500
Digital calipers	[OK] **IN HAND / USED**	Motor, Hall thickness, and ESP32 row-pitch readings recorded on 2026-07-31.
Wire stripper / cutter	[X] NEED TO BUY	~?200
Desoldering pump or wick	[X] OPTIONAL	Useful if you need to lift the TO-92 off the hall module, or fix a bridge. ~?120
Tweezers	[X] OPTIONAL	Strongly recommended for 2mm springs and 1mm linkages. ~?80
Small drill bits (1.5/1.7/2.5mm)	[X] OPTIONAL	FDM holes print 0.2-0.3mm undersize; these clean them out. ~?150

Soldering difficulty - the honest read

Nothing in the demo build requires fine-pitch work. The hardest joint is soldering wires flat onto the ULN2003's 2.54mm header pads - comfortably within basic skill.

The **only** fine-pitch work in the whole project is the custom muscle board (TQFP-32 at 0.8mm pitch, SOIC-16, 0402 passives). That is **beyond basic soldering** - order it assembled (PCBA) or stay on the ULN2003 module. It is not needed for the demo either way.

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5. What to actually spend

Path	Contents	Cost
Cheapest working demo	multimeter, breadboard, solder wire, strippers, USB-C cable	**~?1,100**
+ mechanism	above + springs, homing magnet, fasteners, glue, calipers	**~?2,900**
Everything in this BOM	all of the above + optional tools + consumables	**~?3,900**

Project total if you buy everything: ?935 + ~?3,900 = ~?4,835 of ?15,000. Comfortable.

> **Deliberately excluded: the resin print order.** Do the cheap PETG proving print and make one dot move *before* spending on resin. Every downstream decision rests on a mechanism that has never been physically tested.

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6. Buy these first (this week)

1. **Multimeter** (~?500) - unblocks safe power-up. Nothing else can happen safely without it.
2. **Breadboard + USB-C data cable + solder wire** (~?400) - everything needed for the Tier-0 demo.
3. **Threaded 5.5x2.1mm panel-mount jack** (~?15-?100) - replaces the testing-only inline pigtail in the final pod.

Calipers are now in hand and the motor, Hall thickness, and ESP32 row-pitch measurements are recorded.

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7. Unverified - measure before buying or printing

#	What	Feeds
1	Motor vertical stack re-derivation from the completed owned-part readings	`base_plate`, `mid_plate`, cam bore
2	Exact selected threaded panel-mount jack drawing/link	pod lid jack cradle (currently PLACEHOLDER)
3	Hall module PCB dimensions (or switch to bare TO-92)	`base_plate` hall pocket
4	Actual magnet diameter/thickness from the bill	dock magnet pockets (CAD assumes 8x1mm)

#	What	Feeds
5	ESP32 board length/width if the catalogue envelope proves tight; pin-row pitch is measured 25.6mm	pod board envelope
6	Pogo connector, once chosen	`outer_box` pogo carrier pocket